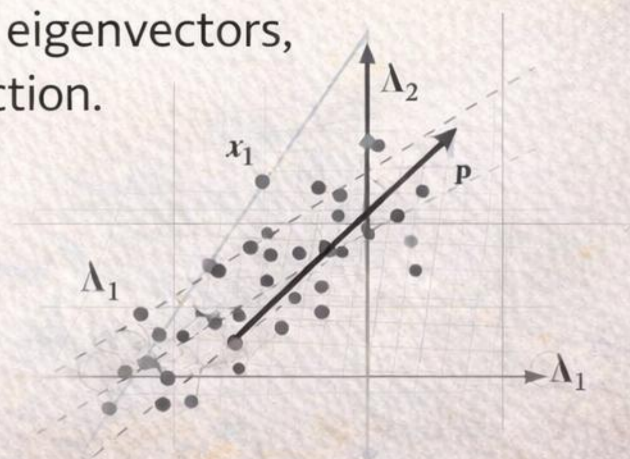
Week 4

Principal Component Analysis

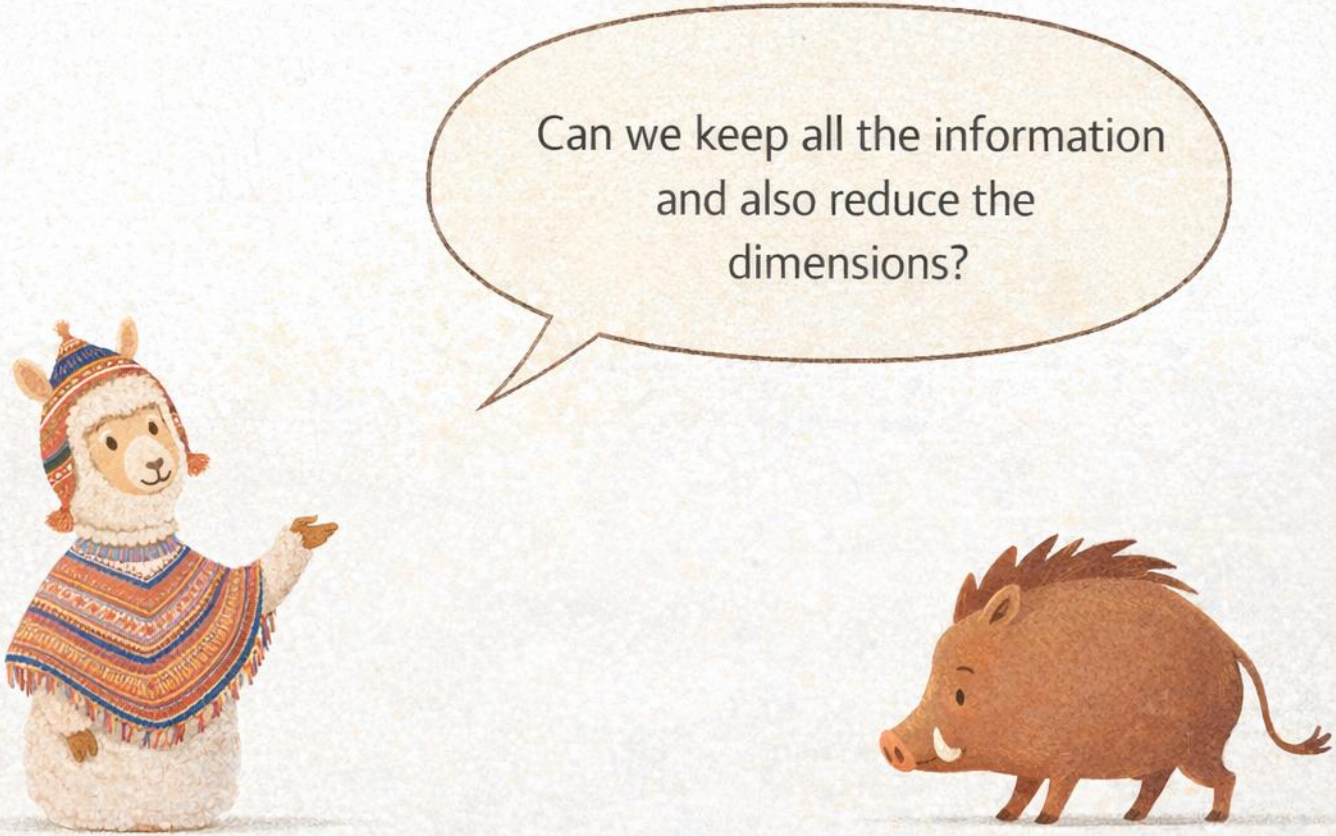
From covariance and projected variance to eigenvectors, orthogonality, and least-squares reconstruction.

$$\Sigma = E\left[(x-\mu)(x-\mu)^{\mathrm{T}}\right]$$



Compression without regret

Mathias wants compression without regret.



PCA starts as an optimization problem

Yunguri reframes PCA correctly.

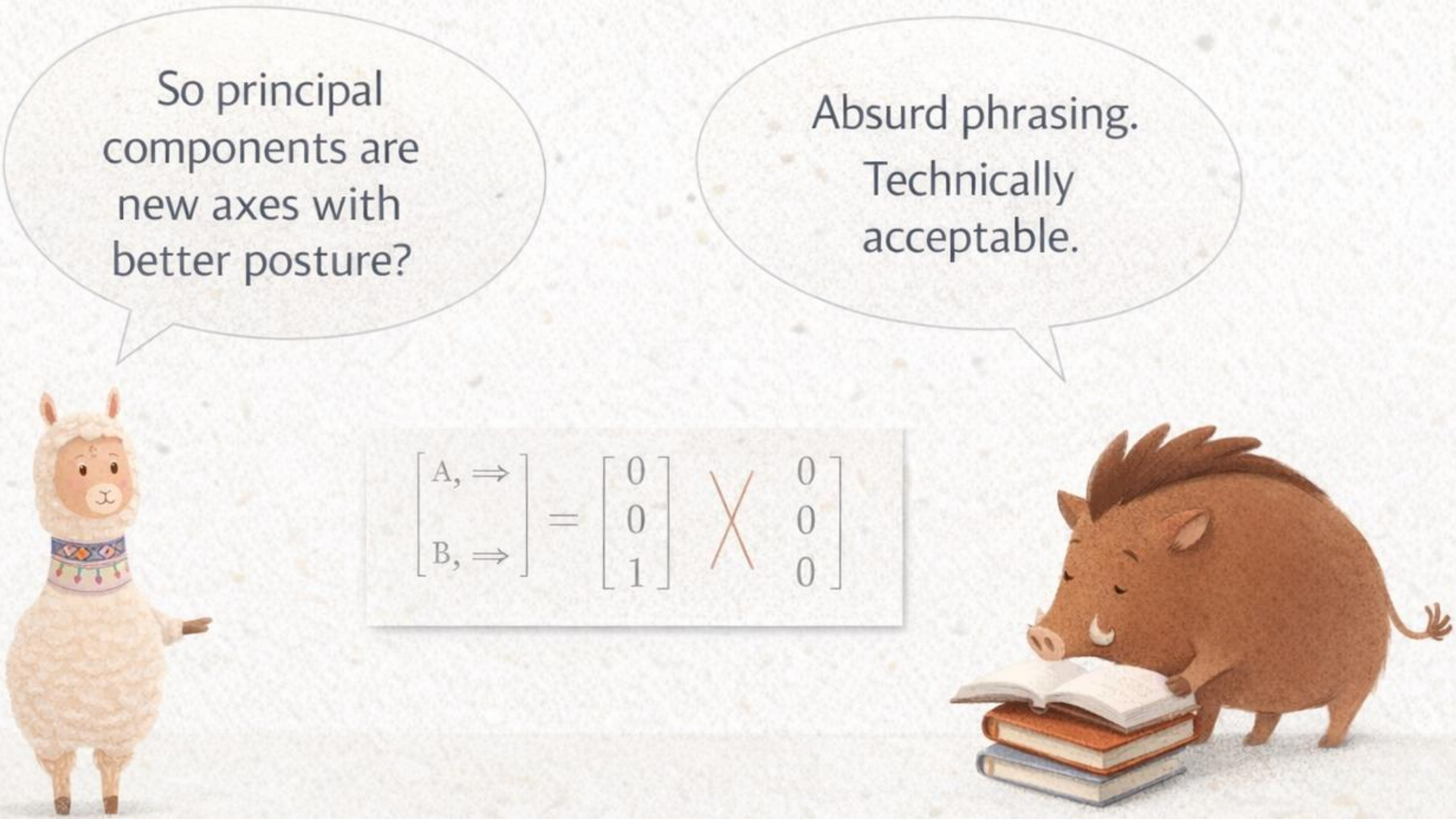
Only if you are ready
to solve a constrained
optimization problem.

$$\text{maximize} \quad Tr(W^T S W)$$

$$\text{subject to} \quad W^T W = I$$

New axes with better posture.

The basis changes; the observations do not.



Where PCA enters the semester

The dataset anchor is the numeric block of pachamix_audio_core.

PachaMix numeric block

family	examples	why PCA cares
rhythm	tempo, danceability	continuous axes invite covariance
spectral	rolloff, centroid, mfcc	many correlated coordinates
timbre	feature bundles	shared structure across tracks
metadata-free	numeric only	ideal for linear algebra

Pipeline: standardize → center → covariance → eigsolve → project

Teaching claim

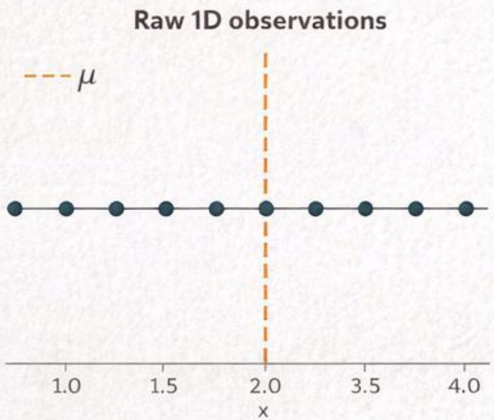
PCA is not introduced as a plotting trick. It is the mathematically principled answer to the question: which k-dimensional linear subspace preserves the most variance?

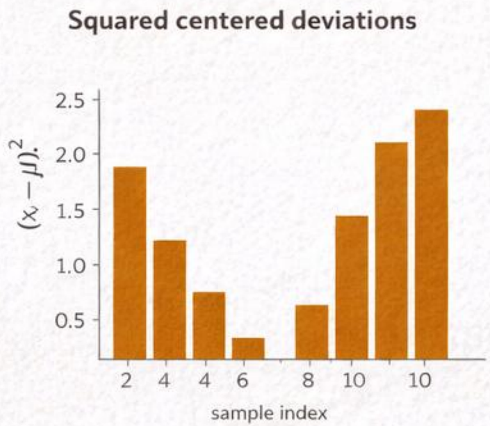
The dataset anchor is the numeric block of pachamix_audio_core. Context: Tς ≐ i1 ⊣ 'i1+z, -z, ƨ, l +1, M, ⊢ X.,

Refined paper-like surfaces, subtie mineral accents, elean hierarchy, and gentle depth while coping *every* @ Ax.

Start with one coordinate: variance of a centered variable

PCA begins with the scalar quantity we already know.





$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\mathrm{Var}(x) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2$$

$$\text{if centered, } \mu = 0 \Rightarrow \mathrm{Var}(x)\ \frac{1}{n}\sum_{i=1}^{n} x_i^2$$

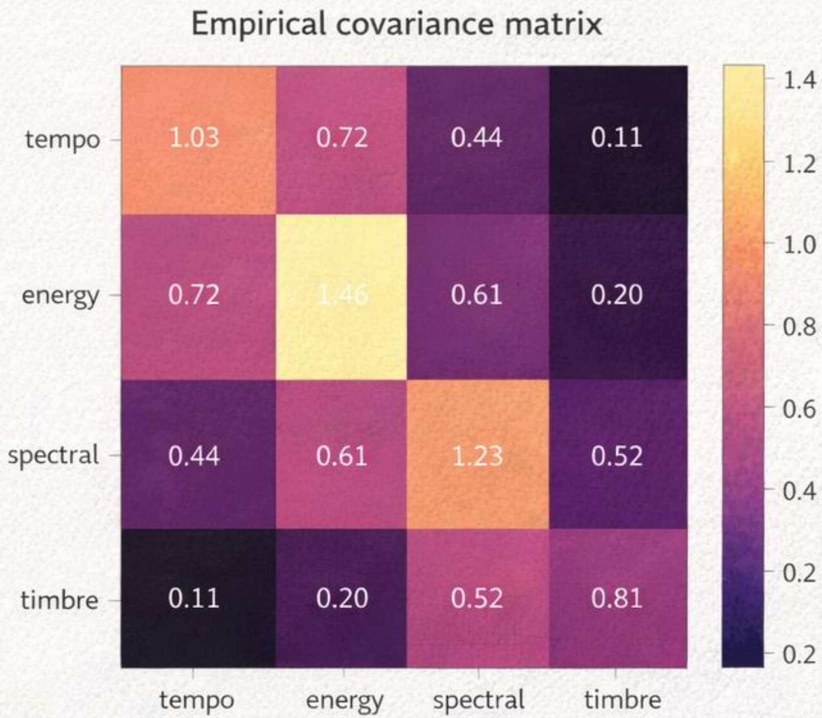
The PCA derivation only makes sense after centering because variance must measure spread around zero.

Key move

Once variance is expressed as a quadratic form, linear algebra becomes the natural language.

From many coordinates to covariance

Covariance records how centered features vary together.



$X \in R^{n} \times d$ columns centered

$$\Sigma = \frac{1}{n} X^{\top} X$$

$$\Sigma_{ab} = \sum_{i=1}^{n} X_{ia} X_{ib}$$

Diagonal terms are feature variances. Off-diagonal terms show how pairs of features co-move.

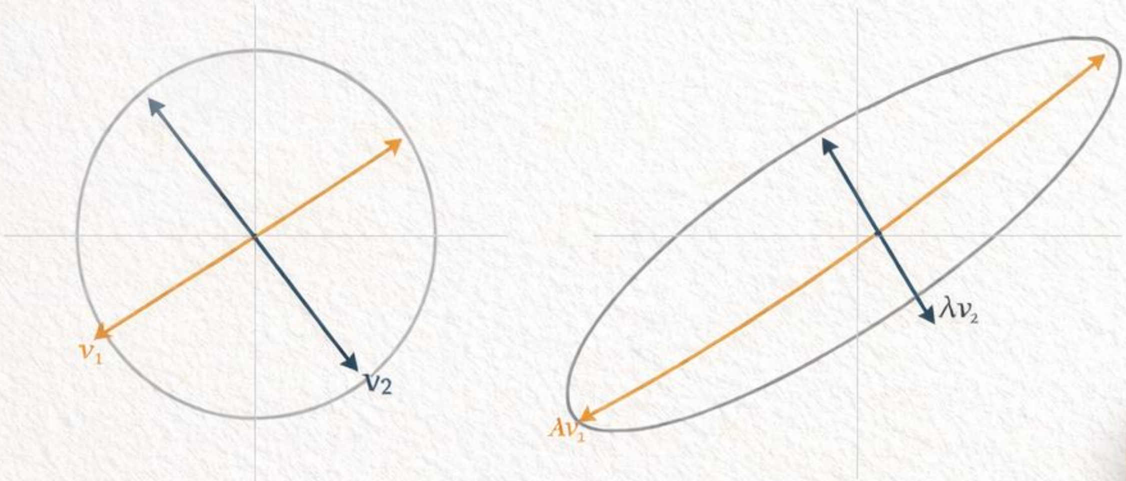
Why this matters

- PCA does not inspect features one by one.
- It studies the second-order geometry of the whole cloud.
- The matrix being diagonalized is built from centered data.

What an eigenvector is before PCA

A matrix can stretch some directions without rotating them away.

Before applying A **After applying A**



$$Av = \lambda v$$

direction preserved up to sign

magnitude scaled by $|\lambda|$

Eigenvectors are the directions a linear operator treats cleanly instead of mixing with others.

Why PCA cares

In PCA the operator is the covariance matrix. We are looking for directions where covariance acts by pure rescaling.

Why covariance gives real, orthogonal directions

ymmetry is what makes PCA mathematically clean.

$$\Sigma^T = \Sigma$$

$$v_i^T \Sigma v_j = \lambda_j v_i^T v_j = \lambda_j v_i^T v_j$$

$$(\lambda_i - \lambda_j)\, v_i^T v_j = 0$$

$$v^T \Sigma v \geq 0 \Rightarrow \lambda_j \geq 0$$

Distinct eigenvalues force orthogonality.
Positive semidefiniteness prevents negative variance directions.

What this buys us

The covariance matrix can be diagonalized in an orthonormal basis, so PCA can rotate the space without distorting angles between retained axes.

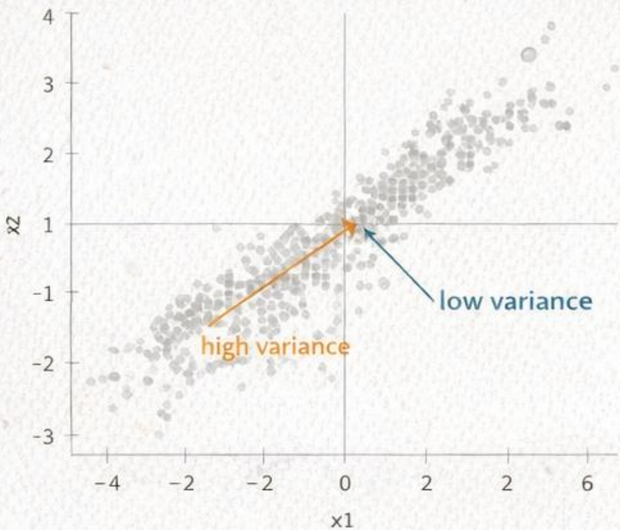
Spectral view

Write Sigma = V Lambda VT. First rotate into the eigenbasis, then read the variance axis by axis from Lambda.

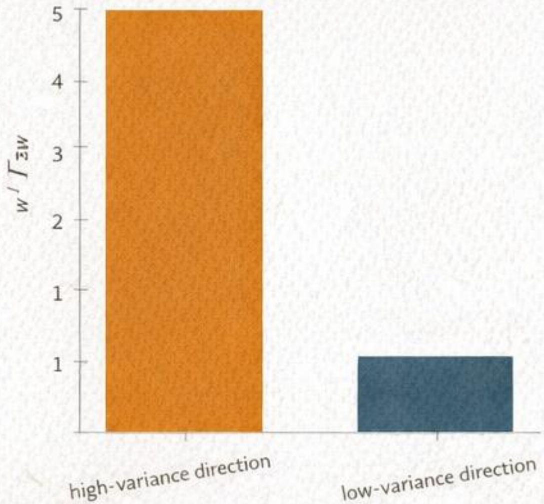
Projected variance depends on direction

Different unit vectors keep different amounts of spread.

One cloud, many candidate directions



Projected variance depends on w



$$z_i = w^T x_i$$

$$\mathrm{Var}(Xw) = \sum_{i=1}^{n} (w^T x_i)^2$$

$$\mathrm{Var}(Xw) = w^T \Sigma w$$

The last identity is the bridge from scalar variance to a matrix optimization problem.

Interpretation

The vector w is not yet special. PCA asks which unit direction maximizes this quadratic form.

The first principal component is an optimization problem

Without a norm constraint, the objective can be scaled arbitrarily.

$$\max_{w \in \mathbb{R}^d} w^T \Sigma w$$

$$\text{subject to } w^T w = 1$$

$$\text{otherwise } (cw)^T \Sigma (cw) = c^2 w^T \Sigma w \rightarrow$$

The unit-norm constraint removes the trivial scaling cheat and leaves only direction as the decision variable.

What is being optimized?

- Input: centered data matrix X.
- Decision variable: a unit direction *w*.
- Objective: maximize variance of the scalar scores Xw.
- Output: the direction that preserves the most spread.

Lagrangian derivation

Differentiate the constrained objective instead of guessing the answer.

$$\mathcal{L}(w, \lambda) = w^{\mathrm{T}} \Sigma w - \lambda(w^{\mathrm{T}} w - 1)$$

$$\nabla w \mathcal{L} = 2\Sigma w - 2\lambda w$$

$$\nabla w \mathcal{L} = 0 \Rightarrow \Sigma w = \lambda w$$

Because Sigma is symmetric, the derivative of $w^{\mathrm{T}} \Sigma w$ is $2\Sigma w$.

$$\nabla w \mathcal{L} = 0 \Rightarrow \Sigma w = \lambda w$$

Why the derivative matters

The optimization problem collapses into a stationary condition. Every critical point of the constrained problem must be an eigenvector of the covariance matrix.

Stationary points satisfy an eigenvalue equation

The candidate directions are exactly the eigenvectors of the covariance matrix.

$$\Sigma w = \lambda w$$

$$w^T \Sigma w = \lambda w^T w$$

$$\|w\|_2 = 1 \Rightarrow w^T \Sigma w = \lambda$$

At any unit eigenvector, the projected variance equals its eigenvalue.

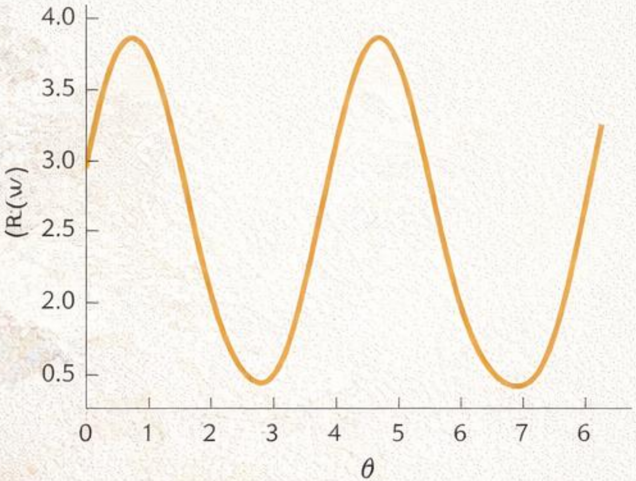
Immediate consequences

- Eigenvectors give admissible stationary directions.
- Eigenvalues measure variance captured along those directions.
- The first principal component is the eigenvector with the largest eigenvalue.

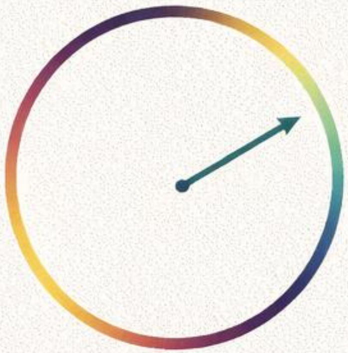
Why the largest eigenvalue wins

The Rayleigh quotient reaches its maximum at the top eigenvector.

Rayleigh quotient over unit directions



The maximizing direction is the top eigenvector



$$R(w) = \frac{wT\Sigma w}{wTw}$$

$$\max_{w \neq 0} R(w) = \lambda_1$$

$$w^* = v_1$$

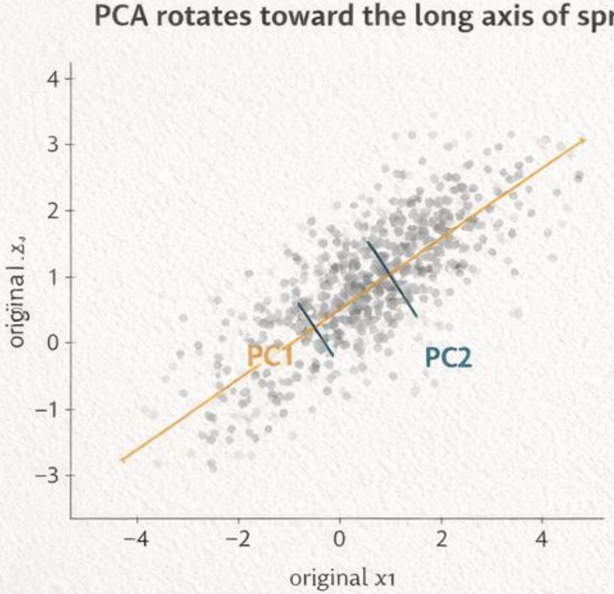
This is the classical extremal property of symmetric matrices, not an ad hoc PCA trick.

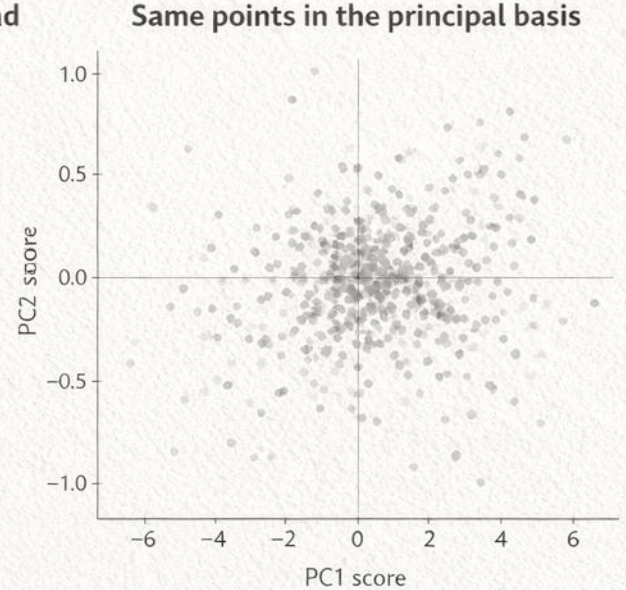
Proof idea

Expand w in the orthonormal eigenbasis of Sigma. The quotient becomes a weighted average of eigenvalues, so it cannot exceed the largest one.

Geometric interpretation: rotate to the longest axis

PCA aligns coordinates with the ellipse of empirical spread.





Geometric reading

- PC1 points along the longest direction of the cloud.
- PC2 is orthogonal and captures the next remaining spread.
- The basis rotates, then the scores become coordinates in that new basis.

Second and later components must be orthogonal

Otherwise the same variance direction would be counted twice.

$$w_1^{\mathrm{T}} w_2 = 0$$

$$W = [w_1, \ldots, w_k] \in \mathrm{R}^{d \times k}$$

$$W^{\mathrm{T}} W = I_k$$

$$\mathrm{Cov}(XW) = W^{\mathrm{T}} \Sigma W = \mathrm{diag}(\lambda_1, \ldots, \lambda_k)$$

Orthogonality prevents repeated counting of the same direction and yields uncorrelated component scores.

Consequences

- Each new component explains fresh variance.
- The score coordinates are decorrelated empirically.
- The ordering follows the spectrum lambda1 >= lambda2 >= ... > lambdad.

Multi-component PCA uses an orthonormal matrix W

The k-dimensional problem is a trace maximization.

$$\max_{W^\top W = I_k} \ \mathrm{tr}(W^\top \Sigma W)$$

$$W^* = [v_1, \ldots, v_k]$$

$$\mathrm{tr}(W^{*\top} \Sigma W^*) = \sum_{i=1}^{k} \lambda_j$$

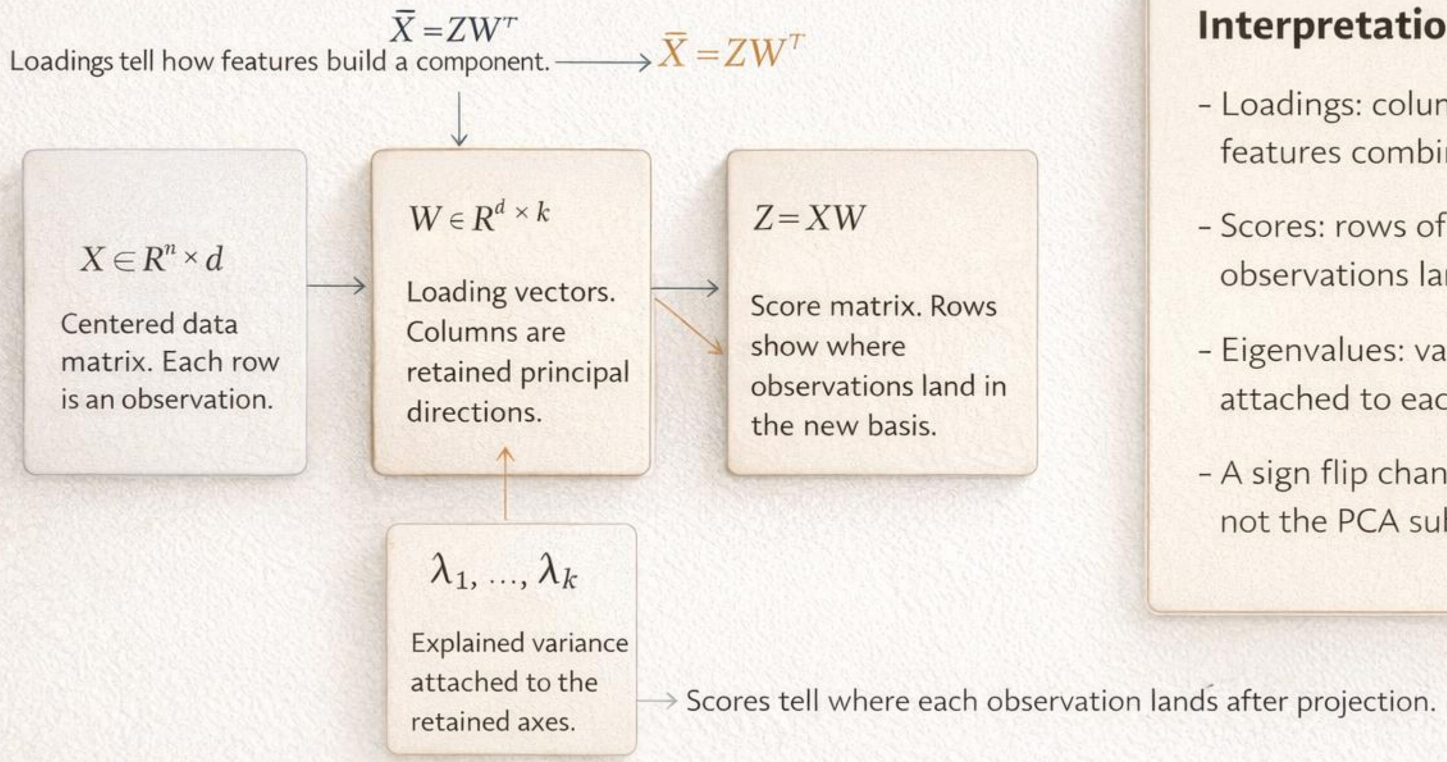
The optimal k-dimensional subspace is spanned by the top k eigenvectors.

Translation to plain language

- Choose k orthonormal axes, not just one.
- Their total preserved variance is the sum of the top k eigenvalues.
- The basis vectors define the projection map $z = W^\top x$

Scores, loadings, and eigenvalues are different objects

Students should name the object before interpreting it.

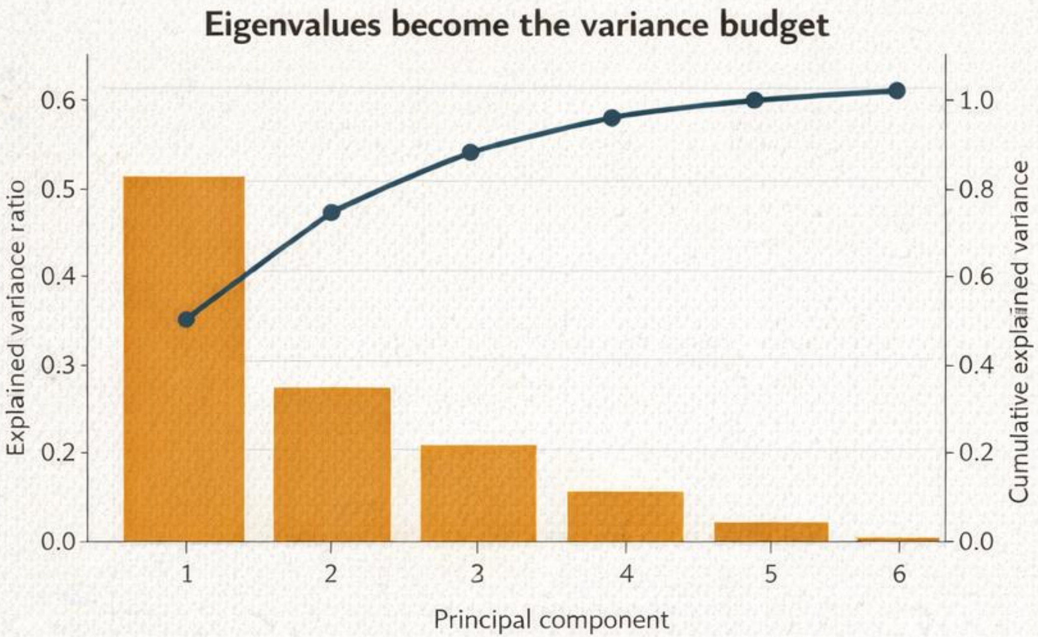


Interpretation guide

- Loadings: columns of W, telling how features combine into a component.
- Scores: rows of Z, telling where observations land after projection.
- Eigenvalues: variance budget attached to each retained axis.
- A sign flip changes orientation, not the PCA subspace itself.

Explained variance is just the spectrum budget

Eigenvalues quantify preserved variance, not meaning.



$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$$

$$\mathrm{EVR}(k) = \frac{\sum^{k} \lambda_j}{\sum_{j=1}^{d} \lambda_j}$$

Interpretation

Explained variance ratio measures second-order variance preserved by the retained subspace.

Common mistake

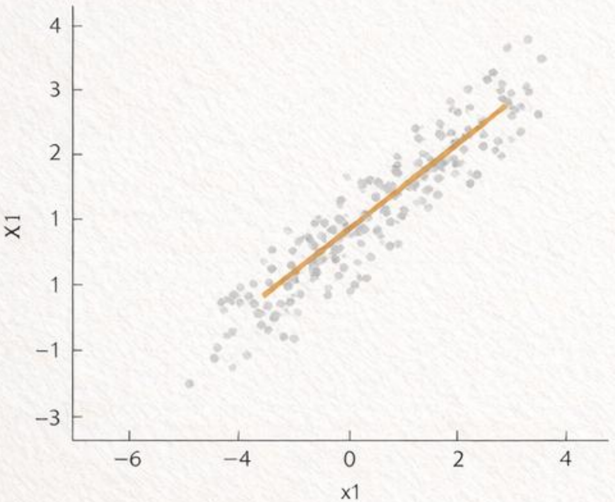
A direction with high variance can still be unhelpful for labels, classes, or interpretation. PCA is unsupervised.

PCA also solves a reconstruction problem

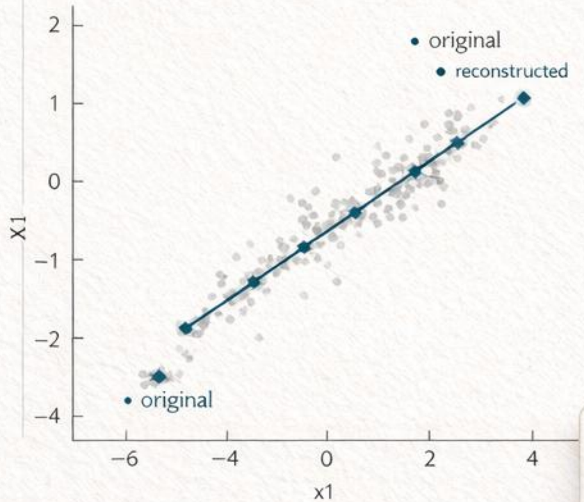
Variance maximization and least-squares projection are equivalent.

$$\hat{X}_i = WW^{\mathsf{T}} x_i$$

Original cloud and the chosen 1D subspace



Orthogonal projection minimizes squared error



$$\min_{W^{\mathsf{T}}W = I} \sum_{k=1}^{n} \left\| \hat{x}_i - W W^{\mathsf{T}} x_i \right\|_2^2$$

$$= \text{const} - \text{n}\; tr\left(W^{\mathsf{T}} \Sigma W\right)$$

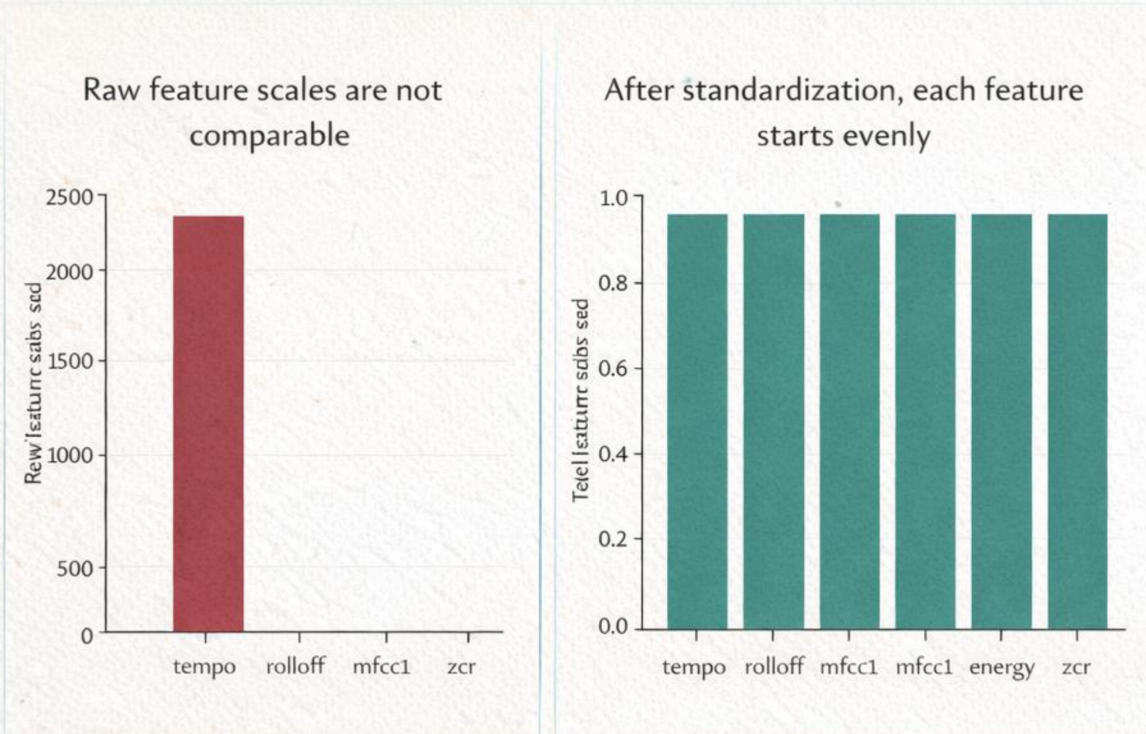
Minimizing projection error is equivalent to maximizing retained variance because the discarded energy appears as a constant minus the trace term.

Dual interpretation

PCA is simultaneously a compression method and an optimal orthogonal projection under squared error.

Standardize or covariance will choose for you

Scale changes the matrix being diagonalized.

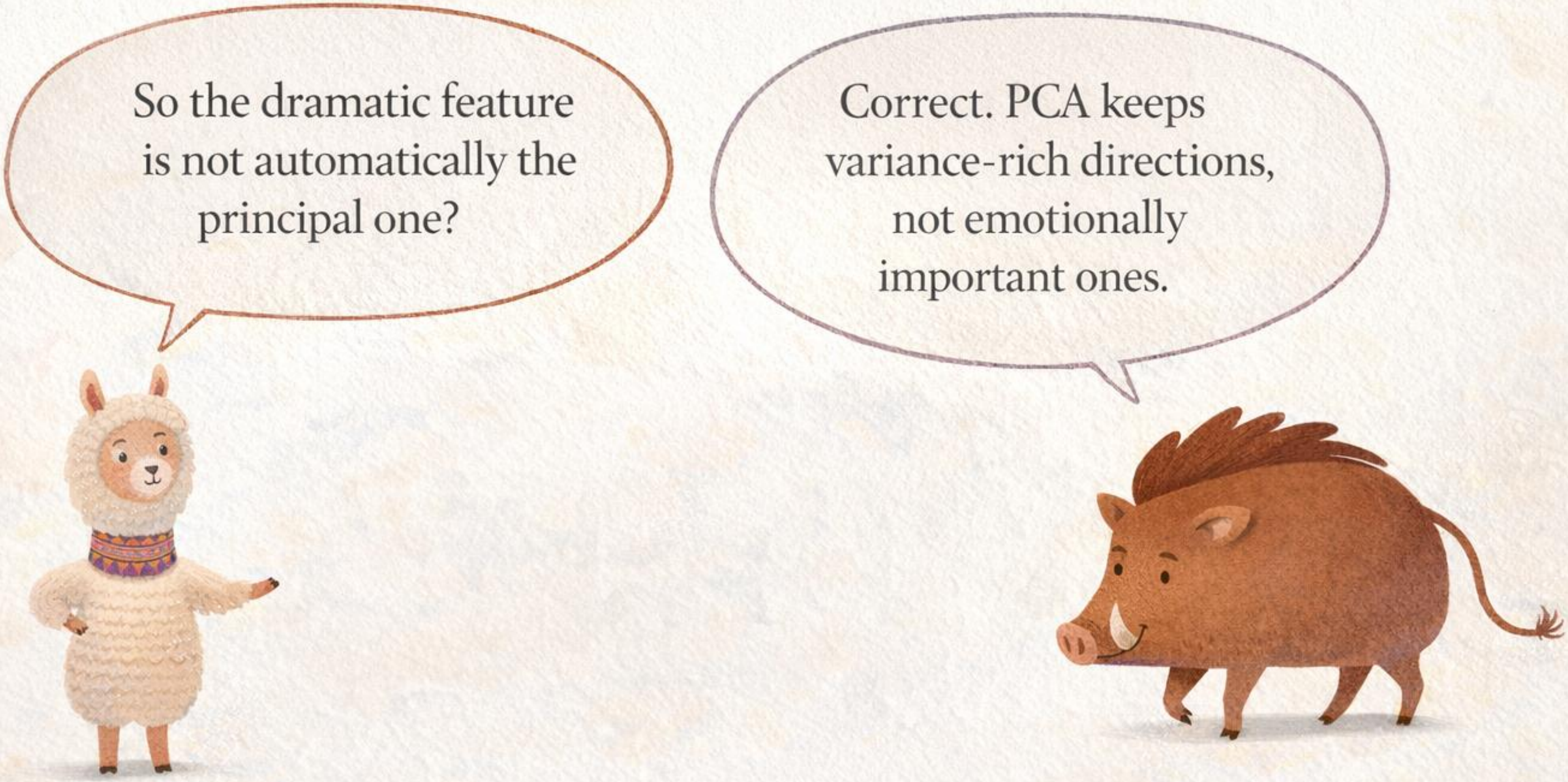


Interpretation warning

- Without standardization, large-scale features dominate Sigma.
- After standardization, PCA behaves more like correlation-based analysis.
- Students must report which version they used before showing any plot.

Variance-rich is not meaning-rich.

Variance preserved is not meaning preserved.



So the dramatic feature is not automatically the principal one? | Correct. PCA keeps variance-rich directions, CENTER

COVARY

PROJECT

Notebook flow for the practical block

Rebuild the derivation numerically on the PachaMix audio matrix.

Execution steps

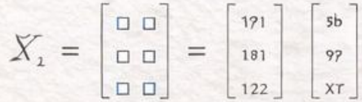
- Load a manageable subset of the numeric audio feature matrix.
- Standardize and center it explicitly.
- Use a toy covariance matrix to inspect eigenvectors before PCA.
- Compute covariance, eigenvalues, and eigenvectors from scratch.

Deliverables and checks

- Inspect orthogonality, residuals, and top feature loadings.
- Project onto PC1, PC2, and several values of k.
- Measure reconstruction error for multiple k values.
- Compare the manual derivation against scikit-learn with sign alignment.

Notebook checkpoints

- Verify numerically that projected variance on PC1 matches the leading eigenvalue.
- Separate scores, loadings, and explained variance before interpreting any result.



Mathematical checkpoint before students leave

These are the statements they must be able to reproduce and defend.

Students must reproduce

- What A v = lamba v means geometrically for covariance.
- The projected-variance identity Var(Xw) = w^T Sigma w.
- The constrained optimization statement for the first component.
- The Lagrangian argument leading to Sigma w = lamba w.
- Why distinct covariance eigenvectors are orthogonal.

Students must not overclaim

- Explained variance is not class separation.
- High variance is not semantic importance.
- Preprocessing choices change the optimization landscape.
- PCA is linear and does not preserve every nonlinear structure.
- Scores, loadings, and eigenvalues are not interchangeable interpretations.

Bridge to Week 5

Week 5 extends the story from eigenspaces to singular values, low-rank approximation, and nonlinear visual embeddings.

Bridge to Week 5

SVD, t-SNE, and embeddings

PCA gave the first principled linear answer. The next week asks what changes when we factor matrices directly, compress by rank, or visualize neighborhoods with methods that are not trying to preserve global variance.

¿Qué es la Reducción de dimensionalidad?

Reducción de dimensionalidad es el proceso de reducir el número de variables de entrada (caracteristicas) en un conjunto de dedatos preservando la máyor cantidad de información